Chapter 16: The Molecular Basis of Inheritance

Complete Master Reference Edition • Fused Curriculum (Introduction, DNA Structure & Replication, Enzymatic Machinery, Telomeres, Lecture Deck Synthesis & Practice Problems)

Course: BSC2010C General Biology I **Institution:** University of Central Florida **Curriculum Block:** Block 4 (Exam 4

Section 16.1: Content Introduction & DNA Chemical Anatomy

Nucleotide architecture, 1' to 5' deoxyribose labeling, purines vs. pyrimidines, phosphodiester linkages, antiparallel double helix geometry, and Chargaff's rules.

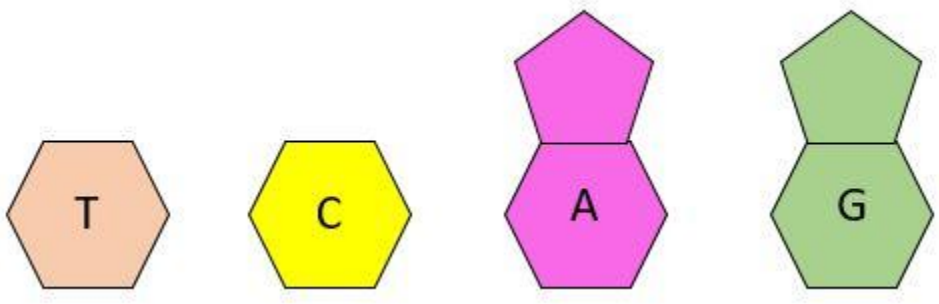
Chapter 16 Content Introduction

Learning Objectives

- Describe the structure of DNA
- Describe prokaryotic versus eukaryotic DNA replication
- Understand the function of enzymes associated with DNA replication
- Compare and contrast leading strand versus lagging strand
- Explain the function of telomeres
- Understand DNA repair

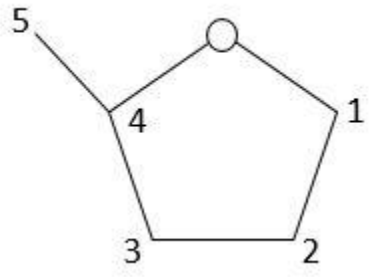
Chapter 16: The Molecular Basis of Inheritance

In this chapter we will learn how DNA replicates itself during S phase of Interphase. Obviously, understanding the structure of DNA is paramount to understanding how it replicates, so we will review DNA structure now. In Chapter 5, we learned that DNA is a double helix made up of nucleotides linked together. Remember that a nucleotide is a pentose sugar attached to a nitrogenous base and to a phosphate group. We also learned that there are only four different nitrogenous bases in DNA: Thymine, Adenine, Cytosine, and Guanine. Thymine and Cytosine are single-ring pyrimidines, Adenine and Guanine are double-ring purines:



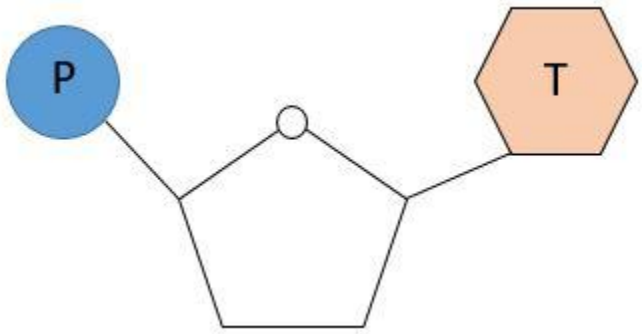
Remember that DNA is anti-parallel and complimentary. Complimentary refers to the base pairing rules: Thymine in one DNA strand always binds with Adenine in the complementary DNA strand, and Cytosine always binds with Guanine. Anti-parallel means that the two strands of DNA run opposite of each other: one runs from 3' to 5' and the other strand runs 5' to 3'.

To assist in understanding the directionality of DNA let's review how we number the carbons in the pentose sugar of DNA. Remember, each "corner" of the ring is a carbon, and there is a terminal carbon that will bond with a phosphate group. We start with the carbon to the right of the oxygen and label it 1, then continue around the ring with the terminal carbon labeled as 5.



Interactive Course Module: Pentose sugar labeling widget

5' carbon bonds to a phosphate group and the nitrogenous base binds to the 1' carbon. Below is an example of this with Thymine as the nitrogenous base:

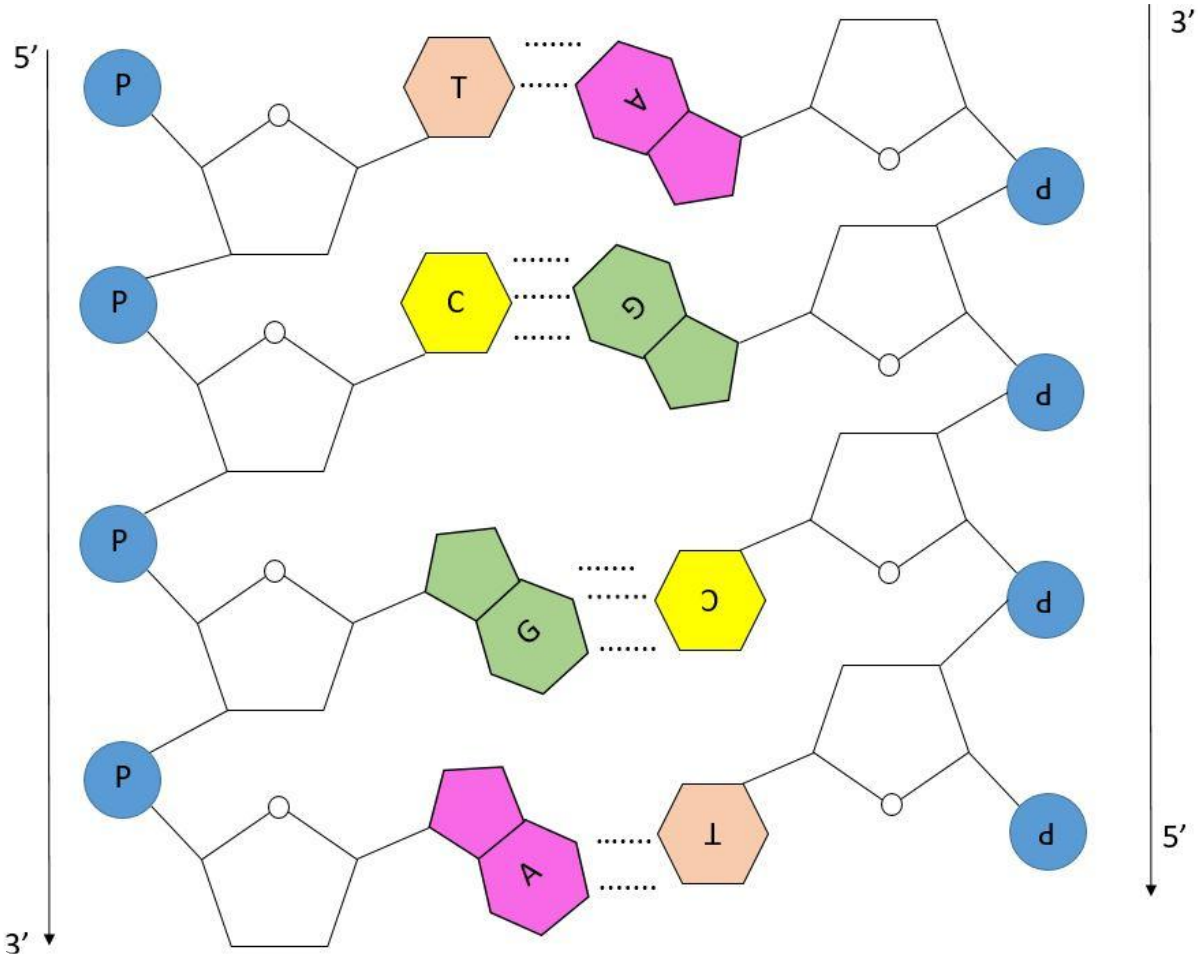


But DNA is not just one nucleotide; it is made up of thousands of nucleotides joined together in a very specific manner. Nucleotides bind to each other via the phosphate group such that the phosphate group attached to the 5' carbon on one nucleotide binds to the 3' carbon of the other nucleotide - this is called a phosphodiester bond. The image below shows just one strand of DNA with four nucleotides - notice that the phosphate group links the nucleotides to each other. You should also take note of the phosphodiester bonds (indicated by the red circles). Take a moment to label the carbons so that you truly understand this - trust me...this is very, very important! The image below is going 5' at the top to 3' at the bottom - the top most carbon is the 5' carbon and the bottom most carbon is the 3' carbon.

Phosphodiester bonds in DNA

Now, let's add the other strand of DNA. Note that is is anti-parallel (runs 3' to 5') and the nitrogenous bases match according to the base pair rules (T-A and C-G). Essentially, one strand runs upside down compared to the other strand; you can see this by noting the orientation of the oxygen in the pentose sugar in one strand

compared to the other. The two strands are held together by hydrogen bonds only! There are two hydrogen bonds between Thymine and Adenine and there are three hydrogen bonds between Cytosine and Guanine.



Interactive Course Module: DNA Structure widget

In addition to understanding the structure of DNA and how all the components are bound together, there are six enzymes and one type of protein that function in DNA replication, you will need to know the name and function of each. We will discuss their specific roles in lecture. They are listed below:

Helicase: unwinds the two DNA strands by temporarily breaking the hydrogen bonds.

Single stranded binding proteins: binds to the newly separated DNA strands to prevent them from re-forming the hydrogen bonds between them.

Topoisomerase: aelieves the overwinding in the DNA double strand caused by helicase's unwinding the two strands.

Primase: attaches a primer of RNA nucleotides that signal for DNA replication to begin at the location of the primer.

DNA Polymerase III (3): adds nucleotides to create new DNA strands using each separated strand as a template.

DNA Polymerase I (1): replaces RNA nucleotide primers with DNA nucleotides.

DNA Ligase: functions in the lagging strand to join the Okazaki fragments into one continuous DNA strand.

Interactive Course Module: Replication Enzymes widget

CHAPTER READING

Chapter 16 textbook reading: DNA Structure, DNA Replication, The Role of Telomeres

CHAPTER LINKS

Chapter 16 PowerPoint

Chapter 16 Post-Quiz

Chapter 16 Practice Problems

Exam 4 Study Guide

Section 16.2: Structural Discovery & The Semiconservative Model

Historical milestones (Hershey-Chase, Chargaff, Franklin, Watson & Crick) and Meselson-Stahl experimental verification of semiconservative replication.

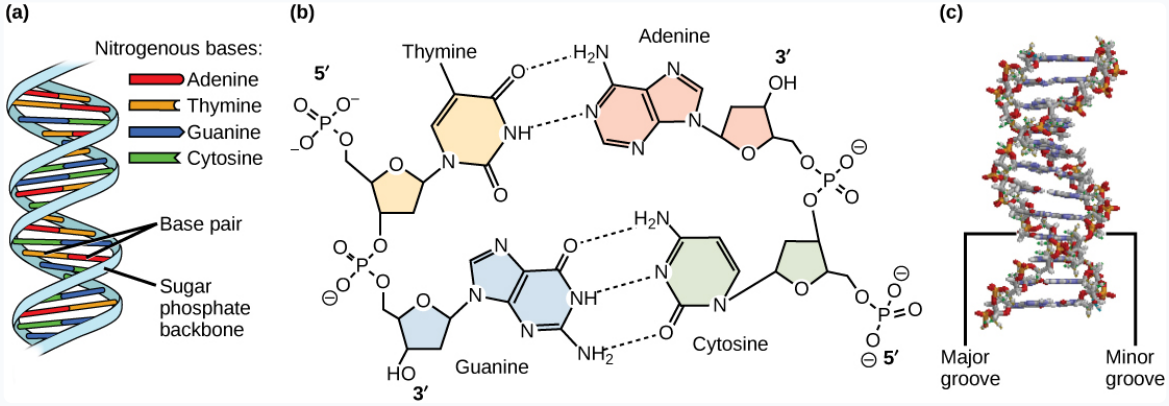
Chapter 16 DNA Structure and Replication Introduction

Learning Objectives

- Describe the structure of DNA
- Explain how the structure of DNA reveals the replication process

Austrian biochemist Erwin Chargaff examined the content of DNA in different species and found that the amounts of adenine, thymine, guanine, and cytosine were not found in equal quantities, and that relative concentrations of the four nucleotide bases varied from species to species, but not within tissues of the same individual or between individuals of the same species. He also discovered something unexpected: That the amount of adenine equaled the amount of thymine, and the amount of cytosine equaled the amount of guanine (that is, A = T and G = C). Different species had equal amounts of *purines* (A+G) and pyrimidines (T + C), but different ratios of A+T to G+C. These observations became known as Chargaff's rules.

Watson and Crick (using Rosalind Franklin's x-ray diffraction images) proposed that DNA is made up of two strands that are twisted around each other to form a right-handed helix. Base pairing takes place between a purine and pyrimidine on opposite strands, so that A pairs with T, and G pairs with C (suggested by Chargaff's Rules). Thus, adenine and thymine are complementary base pairs, and cytosine and guanine are also complementary base pairs. The base pairs are stabilized by hydrogen bonds: *adenine and thymine form two hydrogen bonds and cytosine and guanine form three hydrogen bonds.* The two strands are anti-parallel in nature; that is, the 3' end of one strand faces the 5' end of the other strand. The sugar and phosphate of the nucleotides form the backbone of the structure, whereas the nitrogenous bases are stacked inside, like the rungs of a ladder. Each base pair is separated from the next base pair by a distance of 0.34 nm, and each turn of the helix measures 3.4 nm. Therefore, 10 base pairs are present per turn of the helix. The diameter of the DNA double-helix is 2 nm, and it is uniform throughout. Only the pairing between a purine and pyrimidine and the antiparallel orientation of the two DNA strands can explain the uniform diameter. The twisting of the two strands around each other results in the formation of uniformly spaced major and minor grooves (Figure 14.7).



Part A shows an illustration of a D N A double helix, which has a sugar phosphate backbone on the outside and nitrogenous base pairs on the inside. Part B shows base pairing between thymine and adenine, which form two hydrogen bonds, and be...

Figure 14.7 DNA has (a) a double helix structure and (b) phosphodiester bonds; the dotted lines between Thymine and Adenine and Guanine and Cytosine represent hydrogen bonds. The (c) major and minor grooves are binding sites for DNA binding proteins during processes such as transcription (the copying of RNA from DNA) and replication.

The structure of the double helix provided a hint as to how DNA divides and makes copies of itself. With specific base pairs, the sequence of one DNA strand can be predicted from its complement. The double-helix model suggests that the two strands of the double helix separate during replication, and each strand serves as a template from which the new complementary strand is copied. What was not clear was how the replication took place. There were three models suggested (Figure 14.12): *conservative, semi-conservative, and dispersive.*

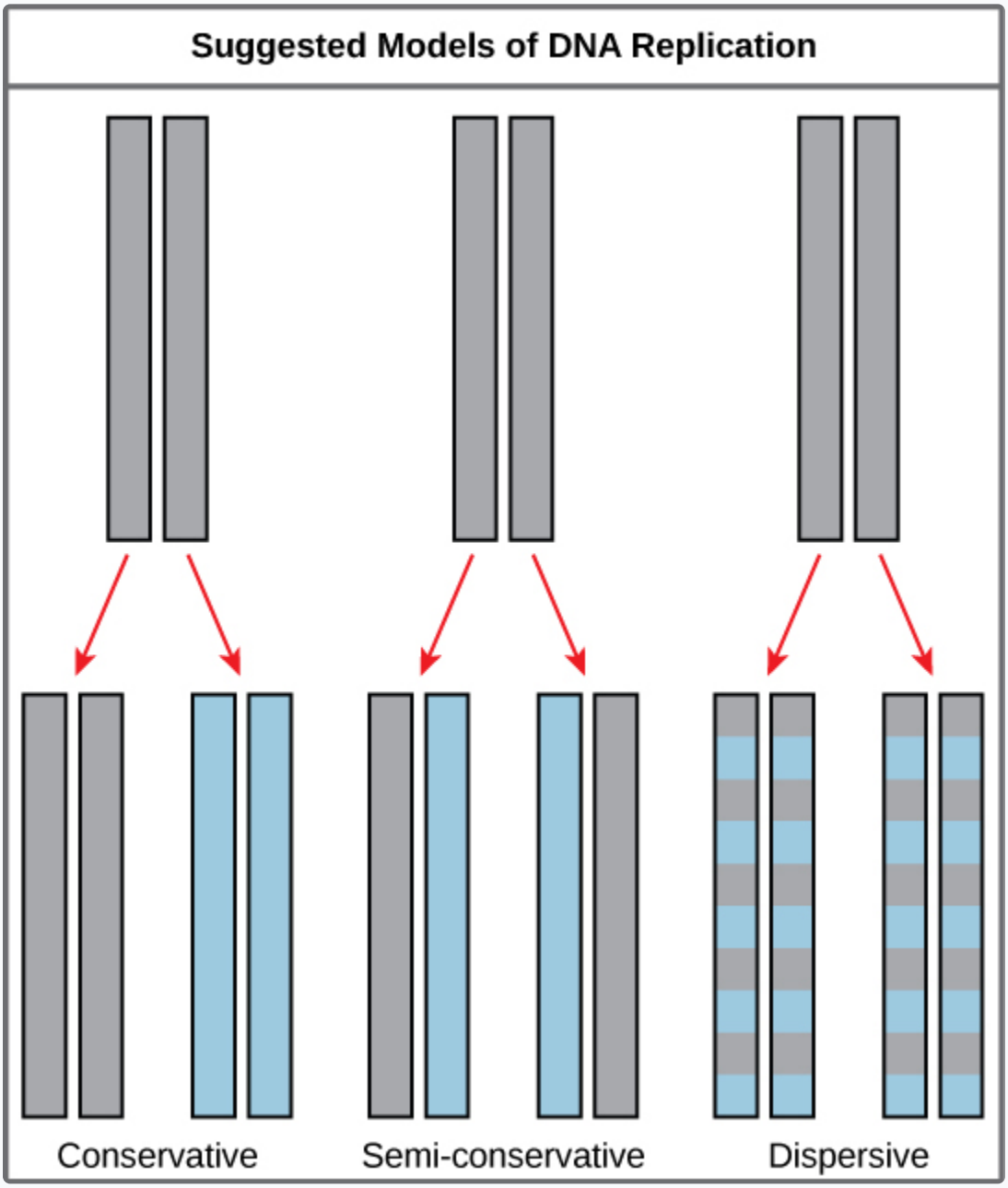


Illustration shows the conservative, semi-conservative, and dispersive models of D N A synthesis. In the conservative model, the D N A is replicated and both newly synthesized strands are paired together. In the semi-conservative model, eac...

Figure 14.12 The three suggested models of DNA replication. Gray indicates the original DNA strands, and blue indicates newly synthesized DNA.

In conservative replication, the parental DNA remains together, and the newly formed daughter strands are together. The semi-conservative method suggests that each of the two parental DNA strands acts as a template for new DNA to be synthesized; after replication, each double-stranded DNA includes one parental or “old” strand and one “new” strand. In the dispersive model, both copies of DNA have double-stranded segments of parental DNA and newly synthesized DNA interspersed.

During DNA replication, each of the two strands that make up the double helix serves as a template from which new strands are copied. The new strands will be complementary to the parental or “old” strands. When two

daughter DNA copies are formed, they have the same sequence and are divided equally into the two daughter cells.

Section 16.3: The Enzymatic Machinery of DNA Replication

Replication origins, bidirectional forks, helicase, single-strand binding proteins, topoisomerase, RNA primase, DNA polymerase III, continuous leading vs. discontinuous lagging strand (Okazaki fragments), DNA polymerase I, and DNA ligase.

Chapter 16 DNA Replication

Learning Objectives

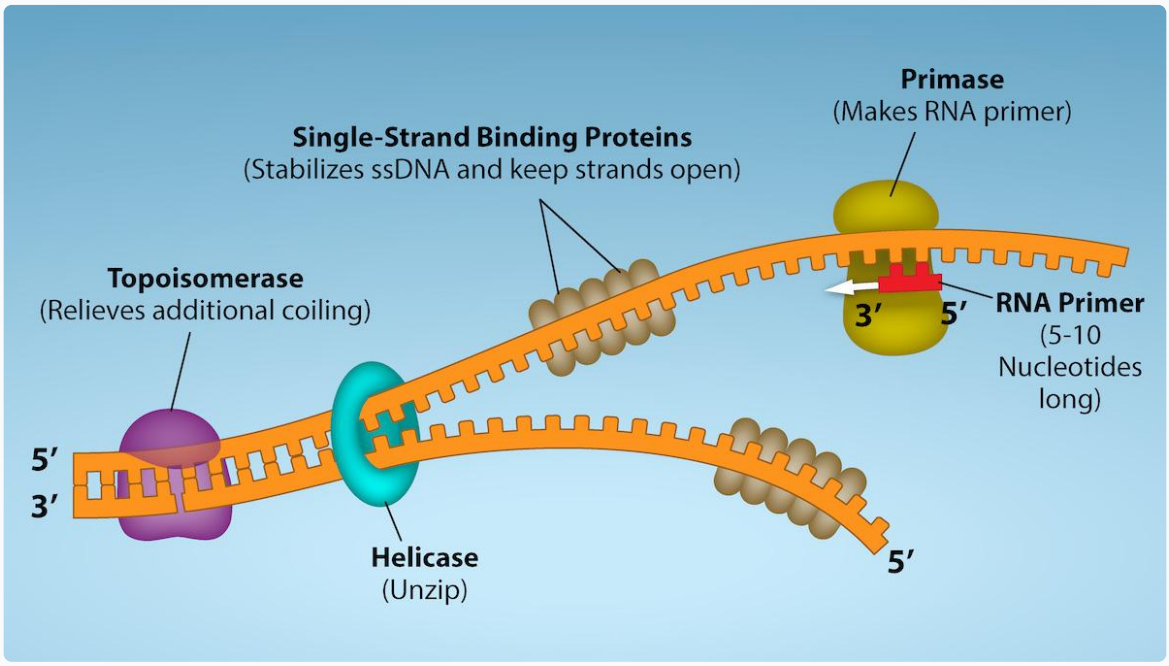
- Explain the process of DNA replication
- Discuss the role of different enzymes and proteins in supporting this process

DNA replication employs a large number of structural proteins and enzymes, each of which plays a critical role during the process. One of the key players is the enzyme **DNA polymerase III** which adds nucleotides one-by-one to the growing DNA chain that is complementary to the template strand. In prokaryotes, three main types of polymerases are known: DNA polymerase I, DNA polymerase II, and DNA polymerase III. **DNA polymerase I** is also an important enzyme in DNA replication. DNA polymerase II is mainly used in repair damaged DNA.

How does the replication machinery know where to begin? It turns out that there are specific nucleotide sequences called *origins of replication* where replication begins. The origin of replication is recognized by certain proteins that bind to this site. An enzyme called ***helicase*** unwinds the DNA by breaking the hydrogen bonds holding the two strands of DNA together. As the DNA opens up, Y-shaped structures called *replication forks* are formed. Two replication forks are formed at the origin of replication and these get extended in each direction as replication proceeds. **Single-strand binding proteins** coat the single strands of DNA near the replication fork to prevent the single-stranded DNA from winding back into a double helix.

DNA polymerase III is only able to add nucleotides in the 5' to 3' direction (a new DNA strand can be only extended in this direction). This means that new nucleotides can only be added to the 3' end of the growing DNA strand. This essentially means that it cannot add nucleotides if the 3' end is not available. Then how does it add the first nucleotide? The problem is solved with the help of a primer that provides the free 3' end. Another enzyme, **Primase**, synthesizes an RNA segment that is about five to ten nucleotides long and complementary to the template DNA. Because this sequence primes the DNA synthesis, it is appropriately called the primer. DNA polymerase III can now extend this RNA primer, adding nucleotides one-by-one that are complementary to the template strand (Figure 14.14).

VISUAL CONNECTION



Topoisomerase binds to D N A to relieve additional coiling. Helicase separates or unzips the two strands. Single-strand binding proteins attach to each D N A strand, and stablize S S D N A and keeps the strands open. While the strands are ...

Figure 14.14 First Components of DNA Replication. As DNA replication begins, DNA Helicase, a large enzyme, separates the two strands of DNA so that they can act as templates for replication. Single-strand binding proteins bind to each strand to stabilize and prevent them from reforming the double helix. Primase, an RNA polymerase, binds to the single stranded DNA and synthesizes a short RNA primer in the 5′ to 3′ direction that is antiparallel to the parental strand. This RNA primer allows for DNA polymerase to begin replicating the DNA. Topoisomerase binds to the double helix upstream of the replication fork to prevent additional coiling by making small cuts in one of the DNA strands. Credit: Rao, A., Ryan, K. Fletcher, S. and Tag, A. Department of Biology, Texas A&M University.

The replication fork moves at the rate of 1000 nucleotides per second. **Topoisomerase** prevents the over-winding of the DNA double helix ahead of the replication fork as the DNA is opening up. Because DNA polymerase III can only extend in the 5' to 3' direction, and because the DNA double helix is *antiparallel*, there is a slight problem at the replication fork. *The two template DNA strands have opposing orientations:* one strand is in the 5' to 3' direction and the other is oriented in the 3' to 5' direction. Only one new DNA strand, the one that is complementary to the 3' to 5' parental DNA strand, can be synthesized continuously towards the replication fork. This continuously synthesized strand is known as the **leading strand**. The other strand, complementary to the 5' to 3' parental DNA, is extended away from the replication fork, in small fragments known as Okazaki fragments, each requiring a primer to start the synthesis. New primer segments are laid down in the direction of the replication fork, but each pointing away from it. Okazaki fragments are named after the Japanese scientist who first discovered them. The strand with the Okazaki fragments is known as the **lagging strand**.

The leading strand can be extended from a single primer, whereas the lagging strand needs a new primer for each of the short Okazaki fragments. As synthesis proceeds, the RNA primers are replaced with DNA nucleotides by **DNA polymerase I**. The enzyme DNA ligase catalyzes the formation of phosphodiester bonds between each Okazaki fragment.

The process of DNA replication can be summarized as follows:

1. DNA unwinds at the origin of replication.
2. Helicase opens up the DNA-forming replication forks; these are extended bidirectionally.
3. Single-strand binding proteins coat the DNA around the replication fork to prevent rewinding of the DNA.
4. Topoisomerase binds at the region ahead of the replication fork to prevent supercoiling.
5. Primase synthesizes RNA primers complementary to the DNA strand.
6. DNA polymerase III starts adding nucleotides to the 3' end of the primer.
7. Elongation of both the lagging and the leading strand continues.
8. RNA primers are replaced with DNA nucleotides by DNA polymerase I.
9. Gaps between the Okazaki fragments are sealed by DNA ligase, which helps in the formation of phosphodiester bonds.

Table 14.1 summarizes the enzymes involved in DNA replication and the functions of each.

Prokaryotic DNA Replication: Enzymes and Their Function

Enzyme/protein	Specific Function
DNA polymerase I	Removes RNA primers and replaces them with DNA nucleotides
DNA polymerase III	Adds nucleotides to the 3' end of the growing new DNA strand
Helicase	Opens the DNA helix by breaking hydrogen bonds between the nitrogenous bases
DNA Ligase	Seals the gaps between the Okazaki fragments to create one continuous DNA strand
Primase	Synthesizes RNA primers needed to start replication
Topoisomerase	Helps relieve the strain on DNA caused by helicase
Single-strand binding proteins (SSB)	Binds to single-stranded DNA to prevent DNA from rewinding back.

Table 14.1

Section 16.4: Linear Chromosome End-Replication & Telomerase

The end-replication problem on the lagging strand, repetitive TTAGGG telomeric non-coding sequences, the Hayflick limit / cellular senescence, and telomerase reverse transcriptase in germline and oncogenic cells.

Chapter 16 The role of Telomeres

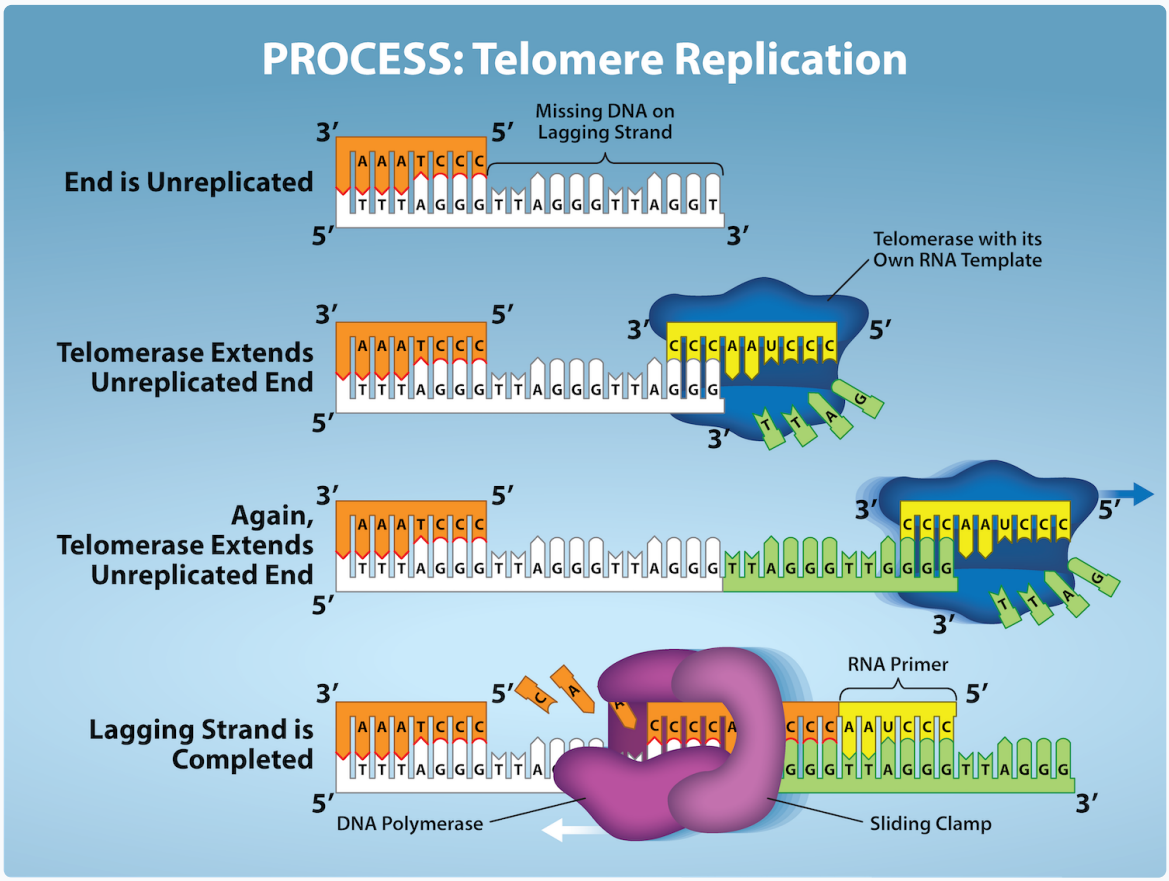
Learning Objectives

- State the role of telomerase in DNA replication

Telomere replication

Unlike prokaryotic chromosomes, eukaryotic chromosomes are linear. As you've learned, the enzyme DNA polymerase III can add nucleotides only in the 5' to 3' direction. In the leading strand, synthesis continues until the end of the chromosome is reached. On the lagging strand, DNA is synthesized in short stretches, each of which is initiated by a separate primer. When the replication fork reaches the end of the linear chromosome, there is no way to replace the primer on the 5′ end of the lagging strand. The DNA at the ends of the chromosome thus remains unpaired, and over time these ends, called **telomeres,** get progressively shorter as cells continue to divide.

Telomeres comprise repetitive sequences that do not code for a gene. In humans, a six-base-pair sequence, TTAGGG, is repeated 100 to 1000 times in the telomere regions. The telomeres protect the genes from getting deleted as cells continue to divide. The telomeres are added to the ends of chromosomes by a separate enzyme, telomerase (Figure 14.16), whose discovery helped in the understanding of how these repetitive chromosome ends are maintained. The telomerase enzyme contains a catalytic part and a built-in RNA template. It attaches to the end of the chromosome, and DNA nucleotides complementary to the RNA template are added on the 3' end of the DNA strand. Once the 3' end of the lagging strand template is sufficiently elongated, DNA polymerase can add the nucleotides complementary to the ends of the chromosomes. Thus, the ends of the chromosomes are replicated. Telomerase is typically active in germ cells and adult stem cells. It is not active in adult somatic cells.



The process of Telomere Replication is shown. The starting image has an unreplicated end, and is missing DNA on the lagging strang. Telomerase has an associated R N A that complements the 5 prime overhang at the end of the chromosome, crea...

Figure 14.15 The ends of linear chromosomes are maintained by the action of the telomerase enzyme. Credit: Rao, A. and Fletcher, S. Department of Biology, Texas A&M University.

Telomerase and Aging

Cells that undergo cell division continue to have their telomeres shortened because most somatic cells do not make telomerase. This essentially means that telomere shortening is associated with aging.

In 2010, scientists found that telomerase can reverse some age-related conditions in mice. This may have potential in regenerative medicine.[2] Telomerase- deficient mice were used in these studies; these mice have tissue atrophy, stem cell depletion, organ system failure, and impaired tissue injury responses. Telomerase reactivation in these mice caused extension of telomeres, reduced DNA damage, reversed neurodegeneration, and improved the function of the testes, spleen, and intestines. Thus, telomere reactivation may have potential for treating age-related diseases in humans.

Cancer is characterized by uncontrolled cell division of abnormal cells. The cells accumulate mutations, proliferate uncontrollably, and can migrate to different parts of the body through a process called metastasis. Scientists have observed that cancerous cells have considerably shortened telomeres and that telomerase is active in these cells. Interestingly, only after the telomeres were shortened in the cancer cells did the telomerase become active. If the action of telomerase in these cells can be inhibited by drugs during cancer therapy, then the cancerous cells could potentially be stopped from further division.

Section 16.5: Instructor Lecture Slide Review (35 Slides Synthesis)

Comprehensive pedagogical synthesis of Dr. Dawson's Chapter 16 lecture presentation (35 slides).

Part A: The Search for the Genetic Material (Slides 1–11)

- **Proteins vs. DNA Historical Debate:**
  - Early biochemists favored proteins as the genetic carrier because proteins are complex polymers of 20 distinct amino acids with versatile conformations, whereas DNA appeared to be a simple, repetitive tetranucleotide polymer of only 4 bases (A, C, T, G).
  - **Phoebus Levene's Tetranucleotide Hypothesis (1940s):** Falsely proposed that DNA consisted of invariant repeating units (AGTC-AGTC...), leading scientists to dismiss DNA as too chemically homogeneous to store complex biological instructions.
- **Landmark Experiments Establishing DNA as the Hereditary Molecule:**
  - **Alfred Hershey & Martha Chase (1952):** Infected *E. coli* bacteria with T2 bacteriophages labeled with radioactive ^{35}S (labeling protein coats) or ^{32}P (labeling phage DNA). After blending and centrifugation, ^{32}P entered bacterial pellets while ^{35}S remained in the supernatant liquid, definitively proving that DNA is the genetic material injected into host cells.
  - **Marie Maynard Daly (1950):** First African-American woman in the United States to earn a Ph.D. in chemistry; developed landmark cell fractionation methods isolating cell nuclei, demonstrated that the four DNA bases (ATCG) varied across species, and conducted foundational research on histone proteins.
  - **Erwin Chargaff (1950):** Analyzed base composition across diverse species and established **Chargaff's Rules**:
    1. DNA base composition varies between species (evidence of molecular diversity).
    2. Within any species, the percentage of Adenine equals Thymine (%A = %T), and the percentage of Guanine equals Cytosine (%G = %C).
  - **Rosalind Franklin (1953):** Expert X-ray crystallographer whose famous "Photo 51" revealed the uniform 2.0 nm diameter, helical pitch (3.4 nm per turn; 10 base pairs per turn), and antiparallel sugar-phosphate backbones facing outward.
  - **James Watson & Francis Crick (1953):** Constructed the correct antiparallel double-helix model. Determined that purines (A, G; double-ring) must pair exclusively with pyrimidines (T, C; single-ring) to maintain uniform 2.0 nm helix width. Purine-purine pairing would be too wide (> 2.0 nm); pyrimidine-pyrimidine pairing would be too narrow (< 2.0 nm).
- **Hydrogen Bonding Specificity:**
  - Adenine pairs with Thymine via **2 hydrogen bonds**.
  - Guanine pairs with Cytosine via **3 hydrogen bonds** (higher thermodynamic stability and denaturation melting temperature).

Part B: The Semiconservative Model & Replication Forks (Slides 12–18)

- **Semiconservative Replication:** Each of the two parental template strands guides the synthesis of a new complementary daughter strand. Each resulting daughter chromosome consists of **one original parental strand and one newly synthesized strand** (demonstrated by Meselson and Stahl using ^{15}N and ^{14}N density-gradient centrifugation).
- **Origins of Replication:**
  - **Prokaryotes (Bacteria):** Single circular chromosome containing **one unique origin of replication** (oriC). Bidirectional replication proceeds outward, forming two replication forks until meeting at the opposite terminus.
  - **Eukaryotes:** Long linear chromosomes containing **hundreds to thousands of replication origins**. Replication bubbles expand bidirectionally and fuse together, dramatically accelerating complete genome replication.
- **The Replication Machinery Battery:**
  - **Helicase:** Enzyme that unwinds and untwists parental double-stranded DNA at replication forks by breaking hydrogen bonds between base pairs.
  - **Single-Strand Binding Proteins (SSBPs):** Monomeric proteins that coat single-stranded template DNA to prevent strands from re-annealing and protect against nucleases.
  - **Topoisomerase (DNA Gyrase):** Relieves torsional overwinding strain ahead of advancing replication forks by nicking, rotating, and resealing phosphodiester backbones.
  - **Primase:** Specialized RNA polymerase that synthesizes short (5–10 nt) complementary RNA primers de novo, providing the essential 3'-OH terminus required by DNA polymerases.
  - **DNA Polymerase III:** Main replicative enzyme in bacteria. Extends new DNA strands in the **5' → 3' direction only**, matching incoming dNTPs to template bases and catalyzing phosphodiester bond formation with pyrophosphate release.

Part C: Leading vs. Lagging Strand Synthesis & Okazaki Fragments (Slides 19–22)

- **The 5' → 3' Directionality Constraint:** Because DNA polymerases can only add nucleotides to a free 3'-OH group, daughter strands can **only grow in the 5' → 3' direction** (reading template 3' → 5').
- **Leading Strand Synthesis:**
  - Synthesized continuously toward the advancing replication fork.
  - Requires only a single initial RNA primer laid down by primase.
  - DNA Polymerase III moves smoothly and uninterruptedly along the template.
- **Lagging Strand Synthesis:**
  - Synthesized discontinuously away from the replication fork.
  - Synthesized in short discrete segments termed **Okazaki fragments** (100–200 nt in eukaryotes; 1000–2000 nt in prokaryotes).
  - Requires multiple RNA primers (one for each Okazaki fragment).
  - **Enzymatic Hand-Off:**
    1. Primase lays down an RNA primer.
    2. DNA Polymerase III extends the fragment until reaching the previous primer.
    3. **DNA Polymerase I** removes RNA primer nucleotides (5' → 3' exonuclease) and replaces them with DNA nucleotides.
    4. **DNA Ligase** seals the remaining phosphodiester nick between adjacent fragments.

Part D: DNA Proofreading, Repair & Telomere Biology (Slides 23–35)

- **Proofreading and Mismatch Repair:**
  - Initial nucleotide misincorporation rate is ~1 in 100,000.
  - DNA Polymerase III has intrinsic **3' → 5' exonuclease proofreading activity**, immediately excising mismatched bases.
  - Post-replicative mismatch repair enzymes scan and correct residual errors, reducing final error rates to **less than 1 in 10 billion nucleotides**!
  - **Nucleotide Excision Repair (NER):** Nuclease cuts out bulky damaged segments (e.g., UV-induced thymine dimers); DNA polymerase fills the gap; DNA ligase seals the backbone.
- **The End-Replication Dilemma of Linear Eukaryotic DNA:**
  - When the terminal RNA primer is removed from the 5' end of the lagging strand, DNA polymerase cannot replace it because there is no upstream 3'-OH primer available.
  - Consequently, each round of eukaryotic DNA replication leaves an un-replicated single-stranded 3' overhang that is degraded, causing chromosomes to shorten progressively.
- **Telomeres and Cellular Aging:**
  - Linear eukaryotic chromosomes terminate in **telomeres**: tandem non-coding repeats (human sequence: **TTAGGG** repeated 100 to 1,000+ times).
  - Telomeres postpone the erosion of essential protein-coding genes near chromosome tips, acting as protective disposable bumpers.
  - Progressive telomere erosion triggers the **Hayflick limit** (cellular senescence / irreversible G0 arrest) or apoptosis, functioning as a tumor-suppressive mechanism in somatic tissues.
- **Telomerase Reverse Transcriptase:**
  - An enzyme consisting of a protein catalytic subunit (TERT) and an internal RNA template (TERC: `3'-AAUCCC-5'`).
  - Elongates the 3' overhanging parental strand by reverse transcription, allowing primase and DNA polymerase to synthesize a final Okazaki fragment, restoring telomere length.
  - Active in germline cells (ensuring zygotes inherit full-length chromosomes), adult stem cells, and **reactivated in ~90% of human malignant cancers**, conferring unlimited replicative immortality.

Section 16.6: Official Course Practice Problems & Verified Solutions

All 19 examination practice problems from Chapter 16 Practice Questions.docx with verified answers and explanations.

1. Which of the following represents a step necessary on the lagging strand, but not the leading strand, during DNA replication?

A) DNA ligase joins Okazaki fragments together (Correct) • B) DNA polymerase synthesizes new DNA strand using parent template • C) Helicase unwinds the parental strand of DNA • D) DNA is synthesized in the 5' to 3' direction

Rationale: The lagging strand is synthesized discontinuously as Okazaki fragments; DNA ligase is required to covalently join these fragments together.

2. What is the function of single-strand binding proteins (SSBPs) during DNA replication?

A) Recruit DNA polymerase • **B) Prevent the separated strands from rebinding (Correct)** • C) Synthesizing RNA primers • D) Ensure both strands are replicated at similar rates

Rationale: Once helicase unwinds the double helix, SSBPs bind to exposed template strands to prevent spontaneous re-annealing into double strands.

3. What is the function of helicase enzyme during DNA replication?

A) Scan for mismatches • B) Add primers to the lagging strand • C) Fix breaks in sugar phosphate backbone • **D) Unwind the double stranded DNA (Correct)**

Rationale: Helicase breaks hydrogen bonds between complementary base pairs at the replication fork, separating the two parental strands.

4. What is the purpose of topoisomerase during DNA replication?

A) Hold DNA polymerases in place • B) Bring new nucleotides to the replication fork • **C) Prevent the DNA from getting overwound as helicase separates the two strands (Correct)** • D) Identify unwanted isomers of nucleotides

Rationale: Topoisomerase relieves supercoiling tension ahead of the replication fork by transiently cleaving and swiveling DNA strands.

5. What would be a direct result of a mutated, nonfunctional gene for primase in a cell?

A) Inability to replicate DNA (Correct) • B) Inability to denature double-stranded DNA • C) Inability to maintain genetic fidelity • D) Inability to transcribe DNA

Rationale: DNA polymerases cannot initiate synthesis de novo; without primase to provide RNA primers, DNA replication cannot begin on either strand.

6. Which of the following refers to the correct model of DNA replication, where each daughter DNA molecule is made up of one parent strand and one newly synthesized strand?

A) Semiconservative (Correct) • B) Dispersive • C) Conservative • D) Random

Rationale: Semiconservative replication, verified by Meselson and Stahl, preserves one intact parental template strand in each daughter duplex.

7. Which of the following proteins are NOT essential to DNA replication?

A) DNA ligase • B) DNA polymerase III • C) Helicase • **D) Cohesin (Correct)**

Rationale: Cohesin holds sister chromatids together during mitosis/meiosis, but is not part of the core replisome machinery synthesizing DNA.

8. T or F. DNA polymerase can only add nucleotides to the 5' end of the DNA strand:

A) True • **B) False (Correct: DNA polymerase adds nucleotides exclusively to the 3'-OH end)**

Rationale: DNA synthesis proceeds strictly in the 5' → 3' direction because nucleotides are attached only to the free 3'-hydroxyl group of an existing nucleotide.

9. Which of the following has the capability of catalyzing the lengthening of DNA in testes cells by adding repetitive sequences of TTAGGG on chromosome ends?

A) Telomerase (Correct) • B) Telomeres • C) Nuclease • D) Primase

Rationale: Telomerase is the reverse transcriptase ribonucleoprotein enzyme that elongates telomeres in germline cells.

10. Which of the following is/are true regarding primers used in DNA replication?

A) RNA polymerase adds RNA nucleotides to new strands • **B) RNA nucleotides are the primers for DNA synthesis (Correct)** • C) Primers are not required on leading strand • D) The primers are protein primers

Rationale: Primers are composed of short RNA oligomers synthesized by primase.

11. According to Chargaff's rule, the following proportion exists in DNA:

A) C = G (Correct) • B) C > T • C) C > G • D) C = T

Rationale: Chargaff's rules state that in double-stranded DNA, %A = %T and %C = %G.

12. In a sample of DNA, 20% is Adenine. How much is Cytosine?

A) 10% • B) 20% • **C) 30% (Correct)** • D) 40%

Rationale: If A = 20%, then T = 20% (total A + T = 40%). The remaining 60% is split equally between G and C: 60% / 2 = 30% Cytosine.

13. Each nucleic acid monomer consists of a sugar, attached phosphate group, and a base. This is called a:

A) Nucleolus • **B) Nucleotide (Correct)** • C) Nucleosome • D) Histone

Rationale: A nucleotide is the tripartite monomer building block of nucleic acids (phosphate + pentose sugar + nitrogenous base).

14. Which statement about the elongation of the lagging strand during DNA replication is correct?

A) It is synthesized in a 3' → 5' direction • **B) It progresses (grows) towards the 3' end of the template strand (away from fork) (Correct)** • C) It requires a short DNA primer to proceed • D) It is synthesized continuously

Rationale: The lagging strand grows in the 5' → 3' direction, reading the template toward its 3' end, which is oriented away from the advancing fork.

15. T or F. The two sides of the double helix structure of DNA are held together by hydrogen bonds:

A) True (Correct) • B) False

Rationale: Nitrogenous bases on opposite antiparallel strands pair through non-covalent hydrogen bonds (A=T with 2 H-bonds, G=C with 3 H-bonds).

16. Which strand grows continuously toward the 5' end of the template strand?

A) Lagging strand • B) DNA strand • C) RNA strand • **D) Leading strand (Correct)**

Rationale: The leading strand is synthesized continuously 5' → 3' into the fork, reading the template toward the template's 5' end.

17. Nucleotides are added to the ____ end of an existing nucleotide:

A) 3' (Correct) • B) 5'

Rationale: DNA polymerase attaches the 5'-phosphate of an incoming dNTP to the free 3'-OH group of the growing strand.

18. How does Prokaryotic DNA differ from Eukaryotic DNA?

A) Prokaryotes have only one chromosome; Eukaryotes have more than one (Correct) • B) Prokaryotes have large quantities of protein-associated DNA • C) Prokaryotes have linear DNA • D) Prokaryotes have many origins of replication

Rationale: Prokaryotes have a single circular chromosome with one replication origin, whereas eukaryotes have multiple linear chromosomes packaged with histones and many replication origins.

19. Which of the following is found at the end of DNA in Eukaryotes?

A) Telomeres (Correct) • B) Topoisomerase • C) Helicase • D) Primase

Rationale: Telomeres cap the physical ends of linear eukaryotic chromosomes.

Appendix: Exam 4 Focus Specifications & Replisome Battery Matrix

Official curriculum requirements from Exam 4 Study Guide 2026.docx covering DNA structure, replication enzymology, and telomere maintenance.

Master Summary Table: The 8 Core Enzymes of DNA Replication & Repair

Enzyme / Protein	Specific Biochemical Function	Strand Association	Consequence of Loss / Mutation
Helicase	Unwinds parental double helix by breaking hydrogen bonds	Replication fork (both strands)	Replication fork cannot open; complete failure to replicate
Single-Strand Binding Proteins	Binds single strands to prevent re-annealing	Template strands ahead of polymerases	DNA strands rebind spontaneously; secondary hairpins halt polymerases
Topoisomerase	Relieves supercoiling strain ahead of fork by transient cleavage	Upstream double helix	Torsional strain halts fork progression; DNA undergoes severe breakage
Primase	Synthesizes short RNA primers with free 3'-OH	Both (1 on leading; many on lagging)	Polymerases cannot initiate; zero DNA synthesis
DNA Polymerase III	5' → 3' elongation of new DNA strands; 3' → 5' proofreading	Leading and lagging strands	Inability to elongate DNA; lethal
DNA Polymerase I	Excision of RNA primers (5' → 3' exonuclease) and replacement with DNA	Both (predominant on lagging)	RNA fragments persist in chromosomes, compromising genetic stability
DNA Ligase	Forms phosphodiester bonds to join Okazaki fragments and repair nicks	Lagging strand predominantly	Unsealed Okazaki fragments; fragmented DNA backbones
Telomerase	Extends 3' overhanging ends of eukaryotic telomeres via reverse transcription	Germline / stem cell chromosome ends	Premature telomere attrition, cellular senescence, gamete genetic loss

Exam 4 High-Yield Conceptual Pitfalls for Chapter 16

- **Directionality Rule:** DNA is ALWAYS synthesized in the 5' → 3' direction. This means the template strand is ALWAYS read in the 3' → 5' direction.
- **Leading vs. Lagging:** The leading strand grows towards the replication fork (continuous). The lagging strand grows away from the fork (discontinuous Okazaki fragments). BOTH strands require primase, but the lagging strand requires multiple primers while the leading strand requires only one.
- **Purine vs. Pyrimidine:** Pure As Gold (Purines = A, G; 2 rings). CUT the Py (Pyrimidines = C, U, T; 1 ring). One purine always pairs with one pyrimidine.